

Contents

IMPORTANT NOTICE: This technical guide and the hardware described are intended solely for Engineering Evaluation and Secondary Development. It is a non-finished sub-assembly. Final product compliance (FCC/ISED) is the responsibility of the integrator.2

1	DOCUMENT REVISION HISTORY.....	2
2	OVERVIEW.....	2
3	SYSTEM CONNECTION DIAGRAM	3
4	HARDWARE DESCRIPTION	3
4.1	UART-WiFi-001.....	3
4.2	SPI-WiFi-002.....	4
4.3	CAN-WiFi-003.....	4
4.4	RS485-WiFi-004.....	5
5	Quick Start with phone setting	5
5.1	Power the board (UART/SPI/CAN/RS485) using the USB interface.....	5
5.2	Connect to the Wi-Fi hotspot.....	6
5.3	Input configuration IP.	6
5.4	Input config parameter.	7
6	Quick Start with Wnet tool config setting.....	9
7	Software Guide	11
7.1	UART-WiFi-001.....	11
7.2	SPI-WiFi-002.....	11
7.2.1	Communication agreement	11
7.2.2	Software design guide.....	12
7.3	CAN-WiFi-003.....	13
7.3.1	Check TCP Status.....	13
7.3.2	CAN and TCP transparent transmission status	13
7.4	RS485-WiFi-004.....	13
8	Reference Link.....	14

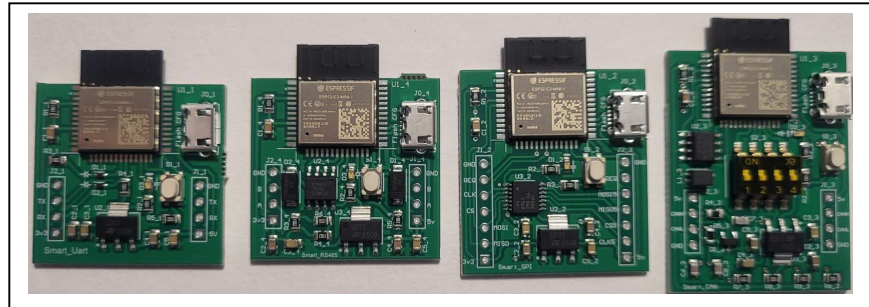
IMPORTANT NOTICE: This Industrial WiFi & Serial Protocol Converter Series is a communication interface component designed for professional system integration and industrial evaluation. It is intended solely for engineering development, demonstration, or secondary evaluation purposes in a controlled laboratory or industrial environment.

1 DOCUMENT REVISION HISTORY

Rev:	Date of issue:	Issued by:
01 First Release	2026-03-18	Hanson Zhang

2 OVERVIEW

- IEEE 802.11b/g/n-compliant
- Center frequency range of operating channel: 2412 ~ 2484 MHz
- Supports 20 MHz, 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Ultra-Low-Power SoC with RISC-V Single-Core CPU
- Evaluation Board Models:
 - UART-WiFi-001
 - SPI-WiFi-002
 - CAN-WiFi-003
 - RS485-WiFi-004



These four conversion boards can be connected to 3.3V or 5V MCUs. For power supply, except for CAN-WiFi-003 which requires 5V, the other three can be powered by 3.3V or 5V. For data communication, these four types are compatible with 3.3V and 5V MCUs and do not require level conversion.

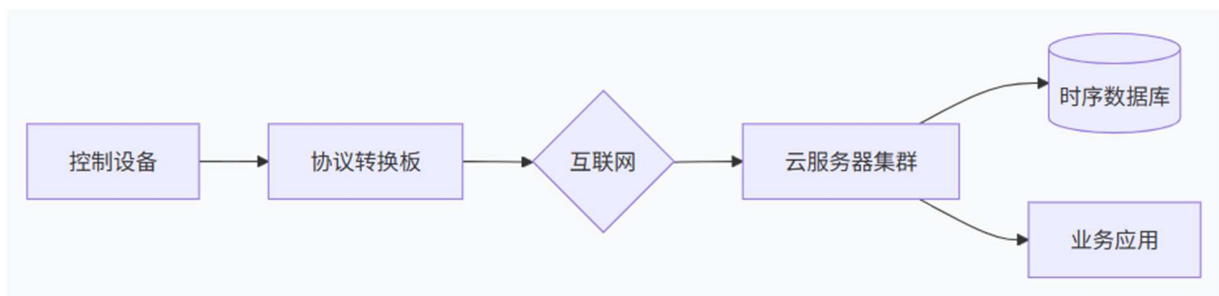
- Bidirectional transparent transmission of data packets between the module and TCP
- FCC Compliance Statement
 - Contains FCC ID: ** 2AC7Z-ESP32C3MINI1
 - Grant Date: ** May 06, 2022
 - Modular Type: ** Single Modular

- Website:

Compliance Documents: <https://github.com/wnettech/compliance>

User Tools: <https://github.com/wnettech/WNetTech-Tool>

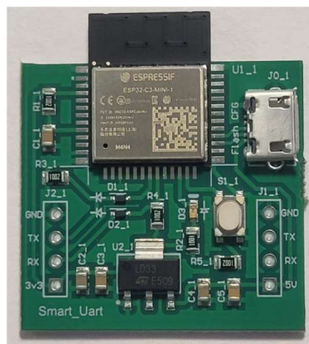
3 SYTEM CONNECTION DIAGRAM



When control equipment needs to add internet communication, our conversion series boards can be used. You can choose the interface board you need according to your requirements, connect to the internet through the interface board, and then connect to the server you need through the internet. After the connection is established, transparent transmission can be achieved. Under transparent transmission, you can define your own protocol to complete different requirements, record data, send/receive control commands, download configuration data and other services.

4 HARDWARE DESCRIPTION

4.1 UART-WiFi-001



Left side connect 3.3v MCU

Right side connect 5v MCU

- * Because RX and TX are compatible with both 3.3V and 5V communication, RX and TX cannot be connected together for loopback testing, must use real MCU TX connect with the module RX, and MCU

RX connect with module TX, you can make loopback software, transmit received data to TX to loopback testing.

4.2 SPI-WiFi-002



Left side connect 3.3v MCU

Right side connect 5v MCU

Interface description:

REQ: interrupt request, low active, the meaning is TCP receive data ready, MCU need to read data from the SPI Evaluation Board.

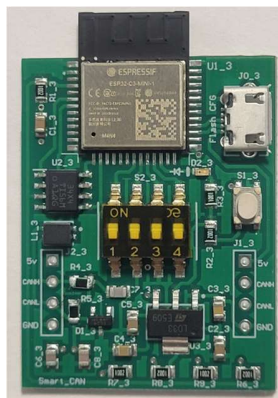
CLK: MCU SPI clk.

CS: MCU SPI chip select, low active.

MOSI: MCU SPI data output, CPI Evaluation Board data input.

MISO: MCU SPI data input, CPI Evaluation Board data output.

4.3 CAN-WiFi-003



Left side Power supply 3.3v

Right side Power supply 5v

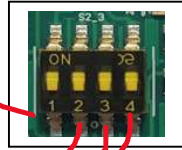
Integrated 120Ω termination resistor for CAN bus.

Power supply is 5v, it has 2 CAN interfaces, connect 2 different MCU CAN interfaces, Generally, it only

connects to one MCU.

Baud rate and CAN ID bit Option:

S2_3-1 ON is 11bits, OFF is 29bits



S2_3-4	S2_3-3	S2_3-2	Baud rate
ON	ON	ON	25K
ON	ON	OFF	50K
ON	OFF	ON	100K
ON	OFF	OFF	125K
OFF	ON	ON	250K
OFF	ON	OFF	500K
OFF	OFF	ON	800K
OFF	OFF	OFF	1M

4.4 RS485-WiFi-004



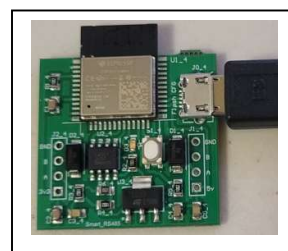
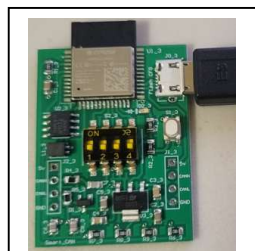
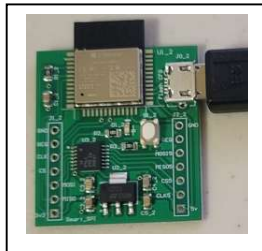
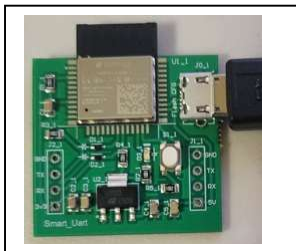
Left side Power supply 3.3v

Right side Power supply 5v.

It has 2 RS485 interfaces, can connect 2 different RS485 devices.

5 Quick Start with phone setting

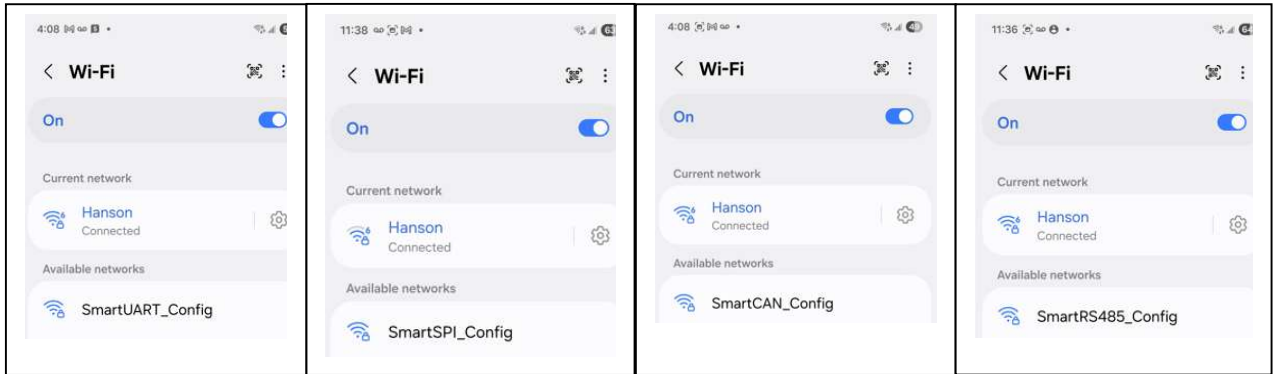
5.1 Power the board (UART/SPI/CAN/RS485) using the USB interface.



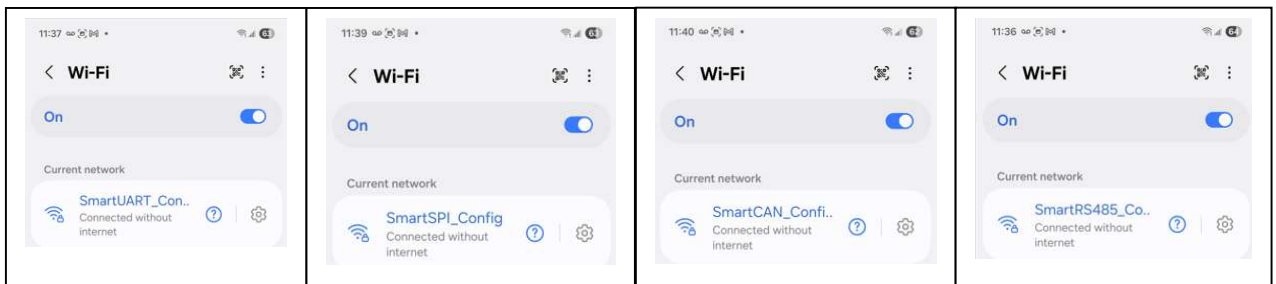
LED will slow flash 1s on, 1s off.

5.2 Connect to the Wi-Fi hotspot.

Open the system settings on your Android phone, find "Connections," and you will see the hotspot device you need. Tap to connect.

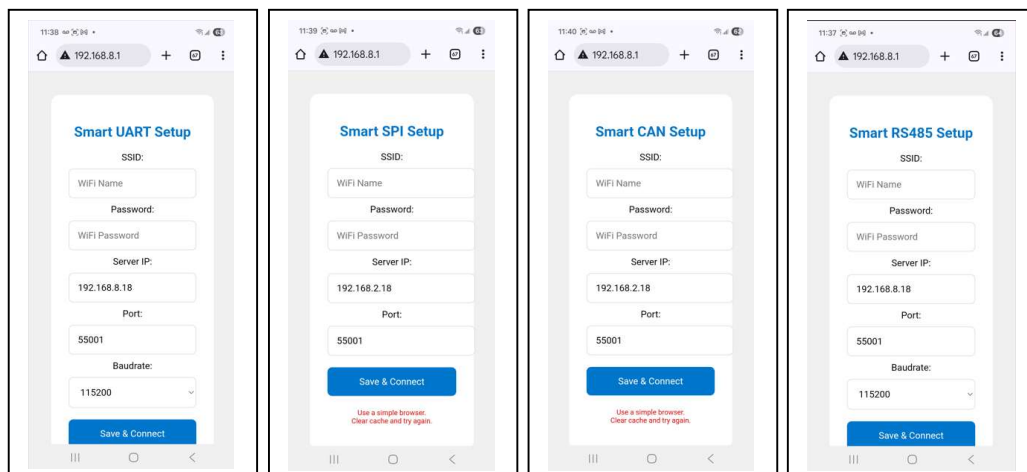
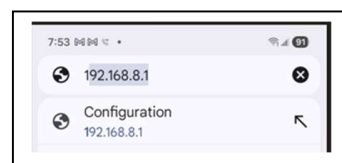


Wait connected



5.3 Input configuration IP.

Using browser input 192.168.8.1 and click



5.4 Input config parameter.

Input your Wi-Fi Name, Wi-Fi Password, Server IP address and Port number, then click Save & Connect,

Cell phones will display:



These parameters will be sent to your series model device, and automatically restart, if your Wi-Fi user ID and password are correct, LED will fast flash 200ms on, 200ms off.

If you want to use server tool test, check your PC IP address:

```

Command Prompt
C:\Users\kevin>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

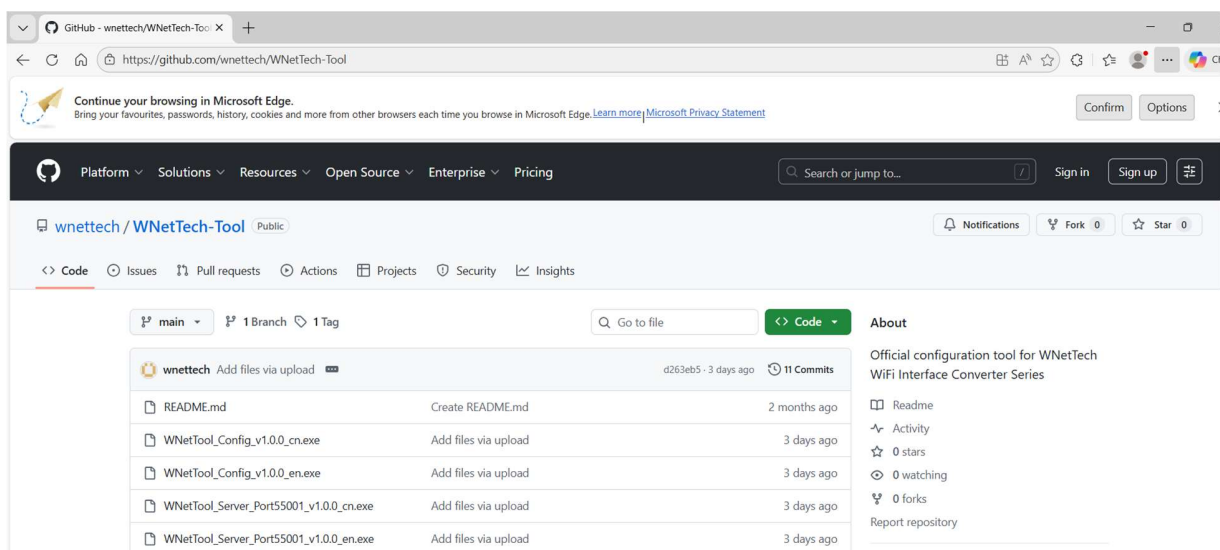
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::7d23:4a72:fd6e:a6a2%8
    IPv4 Address. . . . . : 192.168.1.15
    
```

input your PC IP address 192.168.1.15 and port =55001, click Save & Connect.

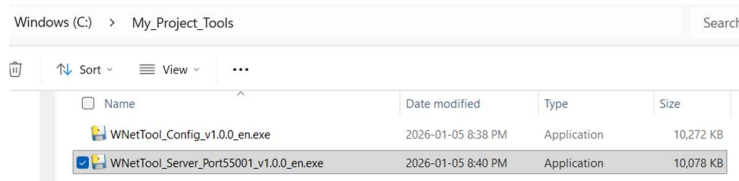
Download server tool, you need to open tool link (with Ctrl+Click):

<https://github.com/wnettech/WNetTech-Tool>



Industrial WiFi & Serial Protocol Converter Series - Technical Guide *Wnet Tech*

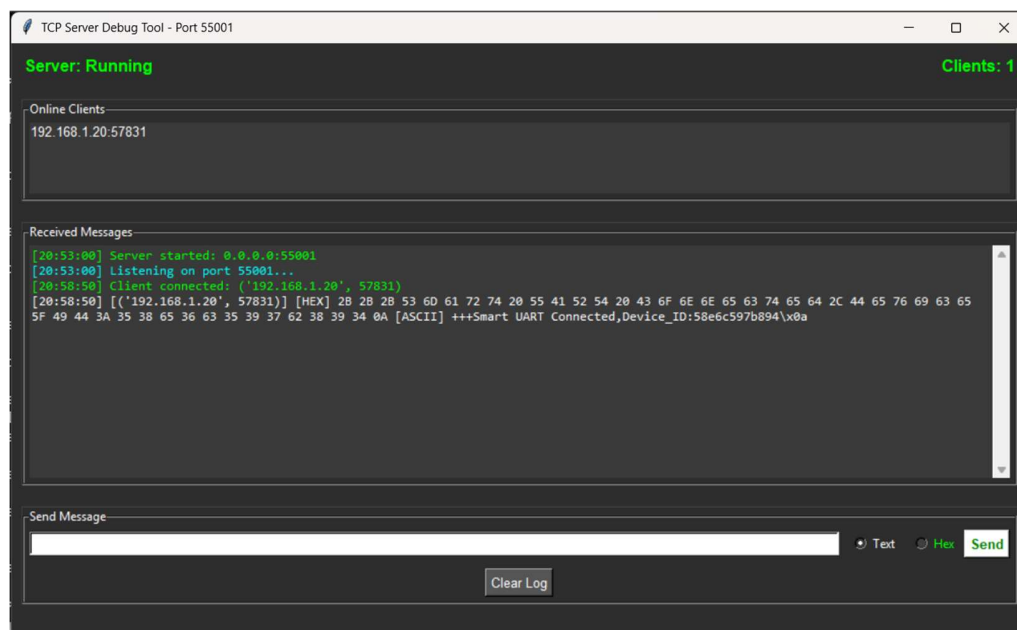
Download WNetTool_Config_v1.0.0_en.exe and WNetTool_Server_Port55001_v1.0.0_en.exe, create folder C: My_Project_Tools, copy these 2 files to this folder.



Double clicks open WNetTool_Server_Port55001_v1.0.0_en.exe,

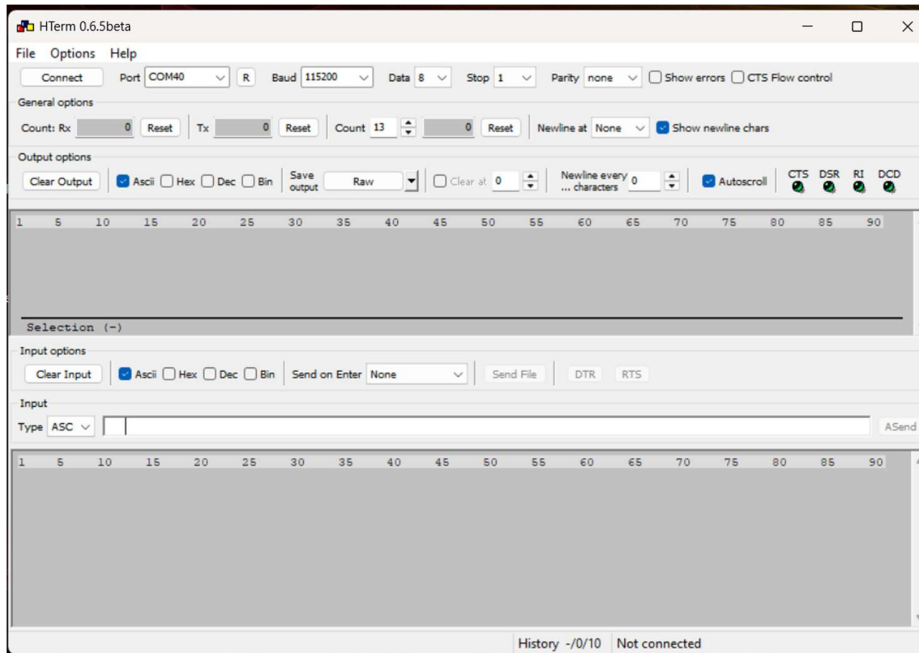


Such as: you use uart, server tool will display:

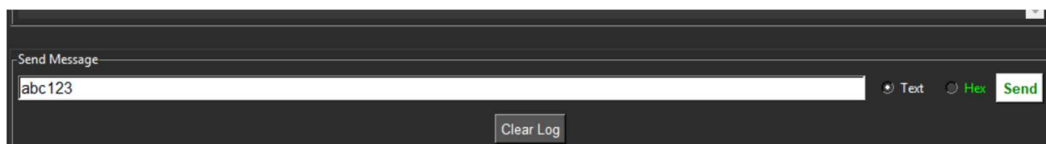


Then, LED will change 200ms flash to 500ms flash, 500ms on, 500ms off.

You can use USB to uart cable, connect PC, find COM number, uart module RX connect with cable TX, uart module TX connect with cable RX, then open any uart tool, such as: HTerm.exe



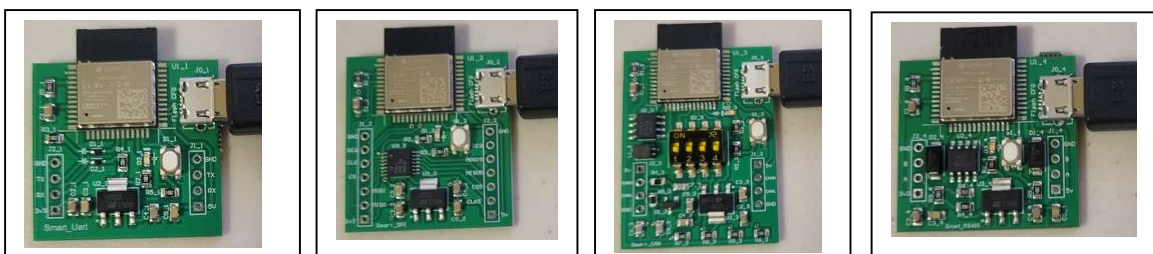
Click connect, input any word or number in server tools ,



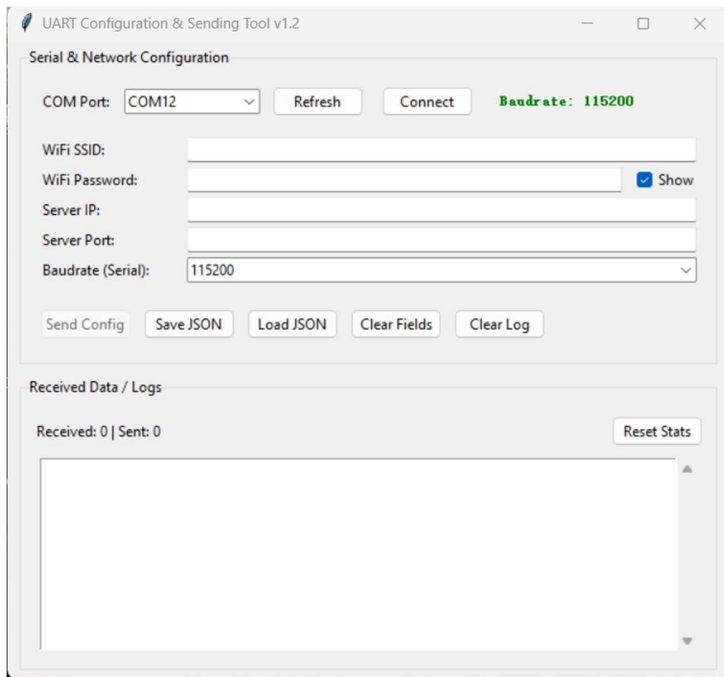
Click send, you will find abc123 to Hterm receive, you also can send data to server with Hterm in same way, realize bidirectional transparent transmission of data between MCU and server with Wi-Fi and internet.

6 Quick Start with Wnet tool config setting

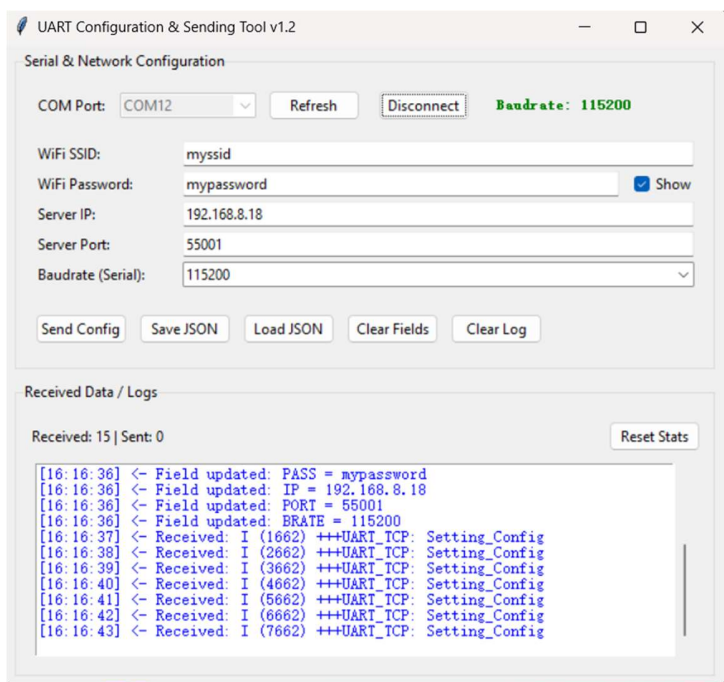
Download WNetTool_Config_v1.0.0_en.exe, power the module (UART/SPI/CAN/RS485) using the USB interface.



Then open file,



Click Connect



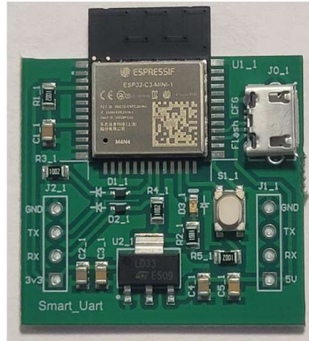
It will automatically read module parameter, WiFi SSID/WiFi Password/Server IP/Server Port, if the module is uart/RS485, also read Baudrate.

Then you can change these parameter, click send Config, these new parameter will send to module, after the module receive these parameter, automatically save these parameter, restart and connect wifi, server.

This tool is good for Batch settings.

7 Software Guide

7.1 UART-WiFi-001



Left side connect 3.3v MCU

Right side connect 5v MCU

MCU TX connect with Evaluation BoardRX, and MCU RX connect with Evaluation BoardTX, after power on, MCU will receive +++info message every 1 second, let you know Evaluation Boardwork status, such as:

a>+++Config_Setting: The meaning is the Evaluation Board not config, it needs setting configuration.

b>+++NK: status corresponds to the LED flashing at 200ms (WiFi Connected).

c>+++OK: status corresponds to the LED flashing at 500ms (Server Connected), bidirectional transparent transmission starts.

MCU can check receive +++OK or more than 1 second no data receive, it means bidirectional transparent transmission starts.

7.2 SPI-WiFi-002



Left side connect 3.3v MCU

Right side connect 5v MCU

7.2.1 Communication agreement

7.2.1.1 Data frame

```
typedef struct {
```

```
    uint16_t header;
```

// 0xAAAA(MCU→WiFi) / 0x5555(WiFi→MCU)/0xB BBBB check receive data length/0xCCCC Check Device status/0xEEEE TCP Disconnect.

```
uint16_t length;    // Data Lenth 0~255

uint8_t data[1460]; // Payload Data

} SPI_Frame;
```

7.2.1.2 Work mode

MCU is Master, Wifi Spi module is slave.

7.2.2 Software design guide

7.2.2.1 module work status

MCU Master SPI can polling send check data to Spi module:

0xCCCC, 0x0002, 0x00, 0x00.(Check Command) every 500ms(or 1s).

Response: 0x0000, 0x0000, 0x00, 0x00, no finding device or device no power.

Response: 0xCCCC, 0x0002, 0x00, 0x00, find device.

Response: 0xCCCC, 0x0002, 'N', 'K', status corresponds to the LED flashing at 200ms (WiFi Connected).

Response: 0xCCCC, 0x0002, 'O', 'K', connect with the server, MCU master should confirm connect, master send confirm data to Spi module: 0xCCCC, 0x0002, 'O', 'K', then go to bidirectional transparent transmission status. status corresponds to the LED flashing at 500ms (Server Connected).

7.2.2.2 Under transparent communication

Send data: 0xAAAA, length, data, MCU master can send data at any time.

Receive data: When receive recv_IRQ interrupt, it stands for SPI module prepare date and wait MCU receive it, MCU master polling send check receive data length,0xB BBBB, 0x0000 every 20ms, polling times set 10, if receive header is 0x5555 or 0xEEEE, exit polling.

If received 0x5555, length, it means MCU need to receive data, then continue send 0xB BBBB, length, 0x00, ..., 0x00, it will receive data,0x5555, length, data.

If receive 0xEEEE,0x0000, it means, TCP disconnect, module need to change status from transparent transmission status to check module work status.

7.3 CAN-WiFi-003



Left side Power supply 3.3v

Right side Power supply 5v

7.3.1 Check TCP Status

Check TCP receive status ID=0x00000002

After power on, receive ID=0x00000002 message, this message polling time is 1s, length=2,

Read Data frame: data[0] = 'N', data[1] = 'K', stands for connecting WiFi or TCP, status corresponds to the LED flashing at 200ms (WiFi Connected).

data[0] = 'O', data[1] = 'K', status corresponds to the LED flashing at 500ms (Server Connected), then go to bidirectional [transparent transmission status](#).

7.3.2 CAN and TCP transparent transmission status

Under transparent transmission, you can send any ID, length and data to CAN module, module will transfer to TCP, data frame is below:

ID0, ID1, ID2, ID3, length, data0, data1, data2, data3, data4, data5, data6, data7, this byte's max is 13, TCP sends data is same frame to CAN module, CAN module will use standard CAN bus sent to MCU.

7.4 RS485-WiFi-004



Left side Power supply 3.3v

Right side Power supply 5v.

Note: RS485 is a half-duplex bus. Ensure the hardware-level auto-direction switching is considered in your timing logic.

MCU A connects with Evaluation BoardA, and MCU B connects with Evaluation BoardB, after power on, MCU will receive +++info message every 1 second, let you know Evaluation Board work status, such as:

a>+++Config_Setting: The meaning is the Evaluation Board not config, it needs setting configuration.

b>+++NK: status corresponds to the LED flashing at 200ms (WiFi Connected).

c>+++OK: status corresponds to the LED flashing at 500ms (Server Connected), bidirectional transparent transmission starts.

MCU can check receive +++OK or more than 1 second no data receive, it means bidirectional transparent transmission starts.

8. Reference Link

For complete compliance certificates (SDoC, FCC ID labels), please refer to our compliance repository: <https://github.com/wnettech/compliance>

9. Compliance Summary

This product series utilizes the pre-certified radio module FCC ID: 2AC7Z-ESP32C3MINI1. All hardware variants (CAN, RS485, SPI, UART) share the identical RF core and power management circuitry. Compliance testing and verification are represented by Model CAN-WiFi-003.